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Study Report of X (same in Y)

AUTODESK

Analysis Title	Lattice Structure - Base Model	Version	Autodesk Fusion (2026.1.0)
Analysis Date	2026-02-27 14:41:19	Solver	Nastran Advanced

Report Properties

Report Properties	
Author	Sunder Ulfarsdt
Author	sunder.ulfarsdt

Simulation Model 1

Study 2 - Static Stress

Study Properties

Study Properties	
Study Type	Linear Stress
Last Modification Date	2026-02-27 14:36:47

Study Setup

Settings		
General		
Analysis Type	3-D solid	
Element Type	Tet	
Mesh		
Average Element Size % of model size	10%	
Curvature Refinement	No	
Max. Number of Elements	50000	
Max. Time Allowed for Meshing (Min.)	60	
Max. Time Allowed for Solve (Min.)	60	
Max. Solver Memory (GB)	8	
Max. Solver Disk (GB)	20	
Solver Precision	Single or Double if necessary	
Material Stiffness Coefficient	0.001	
Adaptive Mesh Refinement	Disabled	
Number of Refinement Steps	2	
Number of Elements to Refine	10	
Percent of Elements to Refine (%)	20	
Resume for Results Refinement	on Mesh Mesh	

Materials		
Assignments		
Unassigned	Strength	
Unassigned	Yield Strength	
Properties		
Density	2.6E-06 kg / mm^3	
Young's Modulus	70000 MPa	
Poisson's Ratio	0.33	
Yield Strength	30000 MPa	
Ultimate Tensile Strength	40000 MPa	
Thermal Conductivity	2.400E-05 W / (mm K)	
Thermal Expansion Coefficient	2.360E-05 / K	
Specific Heat	875.000 J / (kg K)	

Contacts				
Contact	Contact Set	Relationship Type	Enable	Activities
	001 Bondedset1	Symmetric	Master Subsystem Model 1/Body/ Simulation Model 1/...	Flow Activities Heat
	001 Bondset1	Symmetric	Lattice Structure - Base Model 1/Body/ Simulation Model 1/...	Flow Heat Flow Heat

Mesh	
Type	solid
Name	Mesh
Elements	Mesh

Load Case1

Loads	
Mesh	

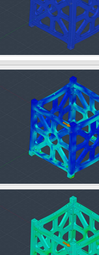
Type	Load
Name	Load
X	Load
Y	Load
Z	Load

Initial Conditions	
Linear Static Load	

Type	Linear Static Load
Acceleration	
X Value	0.00 m / s^2
Y Value	0.00 m / s^2
Z Value	-9.81 m / s^2

Load Case1 - Guided Results

Safety Factor



Will the analysis criteria fit the study, design, the design is expected to fail/underperforming or break

Safety Factor Study Target:

Optimal user notes setup
- Check the load case and set up a target
- Check the Safety Factor distribution and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the displacement result is more than expected, consider running a Nonlinear study to gain more insight

Check other results
- Examine the Stress plot. Would it be satisfied if it was the sole criterion for success and failure?
- Check the Displacement result. Is it acceptable for the intended application?
- Check the Safety Factor plot. Is it acceptable for the intended application?
- Look at other results such as Strain Energy and Strain Energy Density to see which elements are most critical for performance

Strengthen weak areas
- Add mass material to weak areas
- Reduce material in areas with excess strength
- Add fillets
- Modify the shape which is preventing design kinds.

Improve design efficiency
- Add material where there is stress
- Remove material where there is low stress
- Utilize the design
- Reduce the Factor of Safety (FoS) requirements

Next Steps

Safety Factor Study Target:

Optimal user notes setup
- Check the load case and set up a target
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- Remove material where there is low stress
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- Reduce the Factor of Safety (FoS) requirements

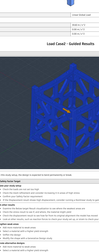
Next Steps

Load Case1 - Results

Result Summary			
Result case	Case results	Minimum	Maximum
Safety Factor			
	Safety Factor	2.25E-006	1.00E+100
Stress			
	Von Mises	-1.00E+100 MPa	3.07E-005 MPa
	1st Principal	-3.07E-005	3.00E-005 MPa
	3rd Principal	-3.00E-005 MPa	3.07E-005 MPa
	Normal X	-3.07E-005	3.07E-005 MPa
	Normal Y	-3.07E-005	3.07E-005 MPa
	Normal Z	-3.07E-005	3.07E-005 MPa
	Shear XY	-3.07E-005 MPa	3.07E-005 MPa
	Shear XZ	-3.07E-005 MPa	3.07E-005 MPa
	Shear YZ	-3.07E-005 MPa	3.07E-005 MPa
Displacement			
	Total	0 mm	8.79E-01 mm
	X	0 mm	8.79E-01 mm
	Y	-8.20E-01 mm	6.20E-01 mm
	Z	-7.60E-01 mm	1.00E-01 mm
Reaction Force			
	Total	0 N	0.00 N
	X	0 N	0.00 N
	Y	-0.00 N	0.00 N
	Z	-0.00 N	0.00 N
Strain			
	Equivalent	-1.05E-08	1.61E-05
	1st Principal	-1.05E-08	1.61E-05
	3rd Principal	-1.16E-05	2.23E-08
	Normal XX	-3.80E-06	3.76E-06
	Normal YY	-1.60E-06	1.85E-06
	Normal ZZ	-1.19E-05	1.48E-05
	Shear XY	-1.05E-05	1.05E-05
	Shear XZ	-4.37E-06	5.00E-06
	Shear YZ	-8.87E-06	1.01E-05
Contact Pressure			
	Total	0.00 MPa	0.00 MPa
	X	-0.00 MPa	0.00 MPa
	Y	-0.00 MPa	0.00 MPa
	Z	-0.00 MPa	0.00 MPa
Contact Force			
	Total	0 N	0.00 N
	X	-0.00 N	0.00 N
	Y	-0.00 N	0.00 N
	Z	-0.00 N	0.00 N

Safety Factor

Safety Factor



Stress

Stress

1st Principal

3rd Principal

Displacement

Total

Load Case2

Loads	
Mesh	

Type	Load
X	Load
Y	Load
Z	Load

Initial Conditions	
Linear Static Load	

Type	Linear Static Load
Acceleration	
X Value	9.81 m / s^2
Y Value	0.00 m / s^2
Z Value	0.00 m / s^2

Load Case2 - Guided Results

Safety Factor

Will the analysis criteria fit the study, design, the design is expected to fail/underperforming or break

Safety Factor Study Target:

Optimal user notes setup
- Check the load case and set up a target
- Check the Safety Factor distribution and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the displacement result is more than expected, consider running a Nonlinear study to gain more insight

Check other results
- Examine the Stress plot. Would it be satisfied if it was the sole criterion for success and failure?
- Check the Displacement result. Is it acceptable for the intended application?
- Check the Safety Factor plot. Is it acceptable for the intended application?
- Look at other results such as Strain Energy and Strain Energy Density to see which elements are most critical for performance

Strengthen weak areas
- Add mass material to weak areas
- Reduce material in areas with excess strength
- Add fillets
- Modify the shape which is preventing design kinds.

Improve design efficiency
- Add material where there is stress
- Remove material where there is low stress
- Utilize the design
- Reduce the Factor of Safety (FoS) requirements

Next Steps

Safety Factor Study Target:

Optimal user notes setup
- Check the load case and set up a target
- Check the Safety Factor distribution and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the displacement result is more than expected, consider running a Nonlinear study to gain more insight

Check other results
- Examine the Stress plot. Would it be satisfied if it was the sole criterion for success and failure?
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Strengthen weak areas
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- Add fillets
- Modify the shape which is preventing design kinds.

Improve design efficiency
- Add material where there is stress
- Remove material where there is low stress
- Utilize the design
- Reduce the Factor of Safety (FoS) requirements

Next Steps

Load Case2 - Results

Result Summary			
Result case	Case results	Minimum	Maximum
Safety Factor			
	Safety Factor	3E8 N/m^2	8.91E+09
Stress			
	Von Mises	-1.00E+100 MPa	3.07E-005
	1st Principal	-3.07E-005	3.00E-005 MPa
	3rd Principal	-3.07E-005	3.07E-005 MPa
	Normal XX	-3.07E-005	3.07E-005 MPa
	Normal YY	-3.07E-005	3.07E-005 MPa
	Normal ZZ	-3.07E-005	3.07E-005 MPa
	Shear XY	-3.07E-005 MPa	3.07E-005 MPa
	Shear XZ	-3.07E-005 MPa	3.07E-005 MPa
	Shear YZ	-3.07E-005 MPa	3.07E-005 MPa

Normal-XY	0.01 MPa	0.05 MPa
Normal-XZ	-0.01 MPa	0.01 MPa
Shear-XY	-0.023 MPa	0.04 MPa
Shear-XZ	-0.018 MPa	0.01 MPa

Displacement

Total	0 mm	0.015 mm
X	-0.01 mm	0.01 mm
Y	-1.24E-06 mm	1.38E-05 mm
Z	-1.88E-06 mm	1.90E-06 mm

Reaction Force

Total	0 N	0.12 N
X	-0.12 N	0.01 N
Y	-0.04 N	0.05 N
Z	-0.07 N	0.04 N

Stress

Equivalent	0.0007 MPa	0.014 MPa
1st Principal	-2.09E-08	0.0147 MPa
2nd Principal	-0.0097 MPa	0.0002 MPa
Normal-XX	-0.012 MPa	0.0095 MPa
Normal-YY	-0.0147 MPa	0.0002 MPa
Normal-ZZ	-0.0002 MPa	0.0147 MPa
Shear-XY	-0.007 MPa	0.0059 MPa
Shear-YZ	-0.0059 MPa	0.007 MPa
Shear-XZ	-0.0002 MPa	0.0002 MPa

Contact Pressure

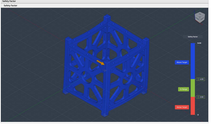
Total	0 MPa	0.07 MPa
X	-0.023 MPa	0.025 MPa
Y	-0.025 MPa	0.025 MPa
Z	-0.02 MPa	0.04 MPa

Contact Force

Total	0 N	0.01 N
X	-0.01 N	0.01 N
Y	-0.01 N	0.01 N
Z	-0.01 N	0.01 N

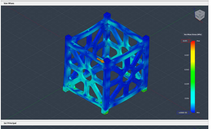
Safety Factor

Safety Factor

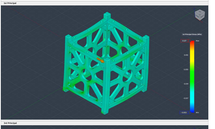


Stress

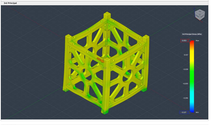
Von Mises



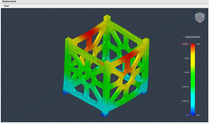
1st Principal



3rd Principal



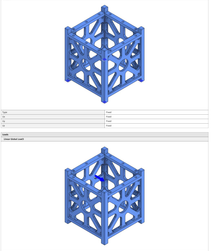
Displacement



Load Case3

Geometry

Assets



Type	Import
Size	1 Part
Time	Static
Time	Static

Loads

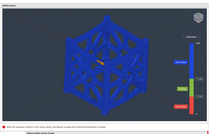
Linear Global Load1

Type	Linear Global Load
Displacement	0 N
Fixed	0.0,0.-1.0,1.0
Free	0.0,0.0,0.0
Center	0.0,0.0,0.0

Load Case3 - Guided Results

Safety Factor

Safety Factor



Warn Design

With this analysis based on the yield strength, the design is expected to hold permanently in steel.

Metric Safety Factor (Yield)

Generate your safety design

- Check the loads are not too high
- Check the boundary conditions are correct (including the contact and other errors)
- Add more supports (fixed, frictionless, etc...)
- Add more contacts (bolted, frictionless, compression, cylindrical, frictionless) to make it a good more rigid

Check other inputs

- Check the material yield strength (tensile/compressive) is as where the material were set
- Ensure the mesh is correct and it is where the material length scale
- Check the material properties are correct (Young's modulus, Poisson's ratio, density, yield strength, ultimate strength)
- Check all other inputs, such as (sensitive) thermal or fluid pressure loads, are as where they are supposed to be

Improve your design

- Add more material to hold stress
- Reduce material where stress is low enough
- Add holes, fillets, etc...
- Add more material where stress is high
- Add more material where stress is low

Metric Safety Factor (Yield)

Generate your safety design

- Check the loads are not too high
- Check the boundary conditions are correct (including the contact and other errors)
- Add more supports (fixed, frictionless, etc...)
- Add more contacts (bolted, frictionless, compression, cylindrical, frictionless) to make it a good more rigid

Check other inputs

- Check the material yield strength (tensile/compressive) is as where the material were set
- Ensure the mesh is correct and it is where the material length scale
- Check the material properties are correct (Young's modulus, Poisson's ratio, density, yield strength, ultimate strength)
- Check all other inputs, such as (sensitive) thermal or fluid pressure loads, are as where they are supposed to be

Lightweight the design

- Add material where stress is high (the design is expected to hold permanently)
- Remove material where stress is low (the design is expected to fail in fatigue, creep, etc...)

Generate alternative designs

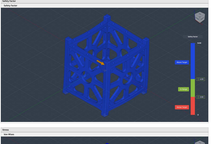
- Generate alternative designs by changing shape, topology optimization
- Generate alternative designs by changing Design Parameters or Sizes

Load Case3 - Results

Result Summary			
Name	Last Value	Minimum	Maximum
Safety Factor	Safety Factor	15.0,0N	15.0,1Min:0N/A
Stress			
Von Mises	0.005 MPa	0.0E-06 MPa	0.015 MPa
1st Principal	-0.08 MPa	-0.08 MPa	0.16 MPa
3rd Principal	-0.08 MPa	-0.16 MPa	0.07 MPa
Normal-XX	-0.003 MPa	-0.008 MPa	0.006 MPa
Normal-YY	-0.006 MPa	-0.016 MPa	0.003 MPa
Normal-ZZ	-0.006 MPa	-0.016 MPa	0.006 MPa
Shear-XY	-0.01 MPa	-0.02 MPa	0.01MPa
Shear-YZ	-0.01 MPa	-0.02 MPa	0.01 MPa
Shear-XZ	-0.01 MPa	-0.02 MPa	0.01 MPa
Displacement			
Total	0 mm	0 mm	0.015 mm
X	0 mm	0 mm	0.015 mm
Y	-1.24E-06 mm	-1.38E-05 mm	1.38E-05 mm
Z	-1.27E-06 mm	-1.90E-06 mm	1.90E-06 mm
Reaction Force			
Total	0 N	0 N	0.12 N
X	-0.05 N	-0 N	0.01 N
Y	-0.05 N	-0.05 N	0.05 N
Z	-0.04 N	-0.04 N	0.04 N
Strain			
Equivalent	1.07E-07 -0M	1.07E-07	0.0E-06
1st Principal	1.07E-07	0.0E-06	1.07E-07
3rd Principal	-1.07E-07	-1.07E-07	0.0E-06
Normal-XX	1.73E07 -0	1.73E07	0.0E-06
Normal-YY	-8.64E07 -0	-8.64E07	0.0E-06
Normal-ZZ	-8.64E07 -0	-8.64E07	0.0E-06
Shear-XY	-4.42E07 -0	-4.42E07	0.0E-06
Shear-YZ	-1.91E07 -0	-1.91E07	0.0E-06
Shear-XZ	-2.03E07 -0	-2.03E07	0.0E-06
Contact Pressure			
Total	0 MPa	0.0E-06 MPa	0.07 MPa
X	-0.025 MPa	-0.025 MPa	0.025 MPa
Y	-0.025 MPa	-0.025 MPa	0.025 MPa
Z	-0.02 MPa	-0.02 MPa	0.04 MPa
Contact Force			
Total	0 N	0 N	0.01 N
X	-0.01N N	-0.01N N	0.01N N
Y	-0.01N N	-0.01N N	0.01N N
Z	-0.01N N	-0.01N N	0.01N N

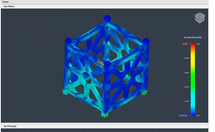
Safety Factor

Safety Factor

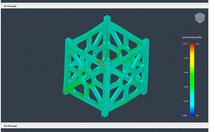


Stress

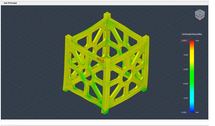
Von Mises



1st Principal

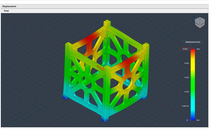


3rd Principal



Displacement

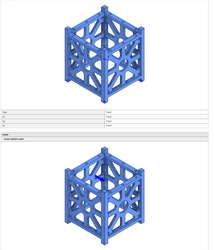
Total



Load Case4

Geometry

Assets



Type	Import
Size	1 Part
Time	Static
Time	Static

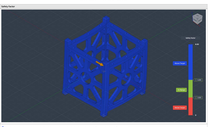
Loads

Linear Global Load1

Type	Linear Global Load
Displacement	0 N
Fixed	0.0,0.-1.0,1.0
Free	0.0,0.0,0.0
Center	0.0,0.0,0.0

Load Case6 - Guided Results

Safety Factor



With the analysis criteria in the setup, there is opportunity to reduce the volume of material and cost of the design.

Next Steps

Meets Safety Factor Limit
- Consider reducing material and cost.
Exceeds safety factor
- Consider ways to reduce stress and improve safety factor, such as:
  - Add fillets and chamfers to reduce stress concentration.
  - Change the geometry so it is more ductile and can absorb more energy before failure.
  - Change the material to a material with higher safety factor, such as:
    - Aluminum Alloy
    - Stainless steel
    - Titanium
    - Carbon fiber composite
    - Fiberglass
    - Plastic
    - Rubber
    - Wood
    - Concrete
    - Ceramic
    - Glass
    - Stone
    - Leather
    - Paper
    - Cardboard
    - Textile
    - Foam
    - Adhesive
    - Sealant
    - Lubricant
    - Coolant
    - Refrigerant
    - Fuel
    - Explosive
    - Propellant
    - Pyrotechnic
    - Biomass
    - Biofuel
    - Biogas
    - Biodiesel
    - Bioethanol
    - Biomethane
    - Biohydrogen
    - Biochar
    - Biocoke
    - Biocrude

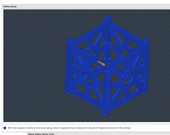
Type	Fixed
Type	Fixed
Type	Fixed
Type	Fixed

Loads

Linear Global Load

Type	Linear Global Load
Coordinate system	Global
Vector	0.0, 0.0, -1.0
X value	0.00 m, 1.0 N
Y value	0.00 m, 1.0 N

Load Case0 - Guided Results



With the analysis criteria in this series, there is opportunity to reduce the volume of material out of the design.

Material Safety Factor Value
- Material Safety Factor Value
- Factor of Safety (FoS) is the ratio of material strength to maximum stress in a part or assembly.
- A FoS of 1 means the material is at its yield point, the material begins yield.
- A FoS of 2 means the material can withstand twice the applied load before reaching the yield strength.

Stress Value
- Stress Value is the amount of force per unit area that is exerted on a body.
- Stress Value is measured in units of pressure, the standard SI unit is pascal (Pa).
- Stress Value is a tensor quantity, it has both magnitude and direction. Stress Value is a tensor quantity.

Displacement Value
- Displacement Value is the amount that a point has moved in space after a load is applied.
- Displacement Value is a vector quantity, it has both magnitude and direction.

Strain Value
- Strain Value is the amount of deformation per unit length that a material has undergone when a load is applied.
- Strain Value is a dimensionless quantity, it has no units.

Contact Pressure Value
- Contact Pressure Value is the amount of force per unit area that is exerted on a body when it comes into contact with another body.
- Contact Pressure Value is measured in units of pressure, the standard SI unit is pascal (Pa).

Contact Force Value
- Contact Force Value is the amount of force that is exerted on a body when it comes into contact with another body.
- Contact Force Value is measured in units of force, the standard SI unit is newton (N).

Safety Factor
- Safety Factor is the ratio of material strength to maximum stress in a part or assembly.
- A Safety Factor of 1 means the material is at its yield point, the material begins yield.
- A Safety Factor of 2 means the material can withstand twice the applied load before reaching the yield strength.

Design Objectives
- Design Objectives are the goals that the software is trying to achieve.
- Design Objectives are based on the Safety Factor, Stress, Displacement, Strain, Contact Pressure and Contact Force.

Contact Analysis Settings
- Contact Analysis Settings are the parameters that define how the software calculates contact between parts in an assembly.
- Contact Analysis Settings include the type of contact (bonded, sliding, etc.), the friction coefficient, and the contact stiffness.

Contact Analysis Results
- Contact Analysis Results are the output of the contact analysis, which includes the contact pressure, contact force, and contact stiffness.
- Contact Analysis Results can be used to identify areas of high contact pressure or force, which may be critical for the design.

Contact Analysis Recommendations
- Contact Analysis Recommendations are suggestions for improving the design based on the contact analysis results.
- Contact Analysis Recommendations may include changes to the contact settings, the geometry, or the material properties.

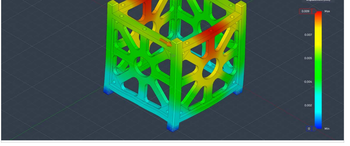
Contact Analysis Warnings
- Contact Analysis Warnings are alerts that indicate potential issues with the contact analysis.
- Contact Analysis Warnings may include warnings about high contact pressure or force, or about contact between incompatible materials.

Contact Analysis Errors
- Contact Analysis Errors are errors that occurred during the contact analysis, which may prevent the analysis from completing.
- Contact Analysis Errors may include errors in the geometry, or in the contact settings.

Contact Analysis Conclusion
- Contact Analysis Conclusion is the overall assessment of the contact analysis, which includes the contact pressure, contact force, and contact stiffness.
- Contact Analysis Conclusion may include a recommendation for improving the design based on the contact analysis results.

Contact Analysis Summary
- Contact Analysis Summary is a brief overview of the contact analysis, which includes the contact pressure, contact force, and contact stiffness.
- Contact Analysis Summary may include a recommendation for improving the design based on the contact analysis results.

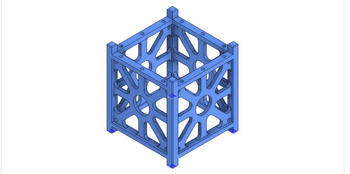
Contact Analysis Report
- Contact Analysis Report is a detailed report of the contact analysis, which includes the contact pressure, contact force, and contact stiffness.
- Contact Analysis Report may include a recommendation for improving the design based on the contact analysis results



Load Case8

Constraints

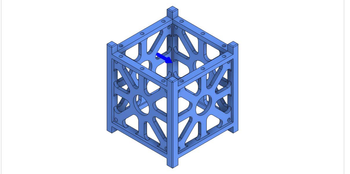
Fixed20



Type	Fixed
Tx	Fixed
Ty	Fixed
Tz	Fixed

Loads

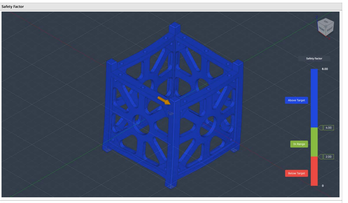
Linear Global Load9



Type	Linear Global Load
Acceleration	
X Value	147.05 m / s^2
Y Value	0.00 m / s^2
Z Value	0.00 m / s^2

Load Case8 - Guided Results

Safety Factor



With the analysis criteria in this study setup, the design is expected to bend permanently or break.

Next Steps

Below Safety Factor Target

Validate your study setup

- Check the loads are not too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results

- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Strengthen weak areas

- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Generate alternative designs

- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Above Safety Factor Limit

Validate your study setup

- Check the loads are not too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results

- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Lightweight the design

- Set a Safety Factor upper limit to find where material can be removed
- Remove material from areas that exceed the Safety Factor upper limit

Generate alternative designs

- In the Generative Design workspace, create design modifications
- Remove unnecessary material using a Shape Optimization study

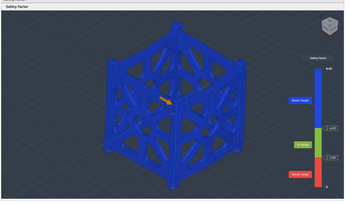
Load Case8 - Results

Result Summary

Result type	Sub result	Minimum	Maximum
Safety Factor			
	Safety Factor	74.606	110,676.805
Stress			
	Von Mises	3.614E-04 MPa	0.536 MPa
	1st Principal	-0.398 MPa	0.952 MPa
	3rd Principal	-0.954 MPa	0.397 MPa
	Normal XX	-0.435 MPa	0.416 MPa
	Normal YY	-0.440 MPa	0.444 MPa
	Normal ZZ	-0.93 MPa	0.93 MPa
	Shear XY	-0.049 MPa	0.047 MPa
	Shear YZ	-0.098 MPa	0.101 MPa
	Shear XZ	-0.138 MPa	0.171 MPa
Displacement			
	Total	0 mm	0.053 mm
	X	0 mm	0.053 mm
	Y	-9.527E-04 mm	9.617E-04 mm
	Z	-0.001 mm	0.001 mm
Reaction Force			
	Total	0 N	2.537 N
	X	-0.214 N	0.073 N
	Y	-0.298 N	0.373 N
	Z	-2.381 N	2.371 N
Strain			
	Equivalent	1.937E-07	2.285E-04
	1st Principal	9.232E-08	2.595E-04
	3rd Principal	-2.602E-04	-9.294E-08
	Normal XX	-8.788E-05	8.689E-05
	Normal YY	-4.325E-05	4.291E-05
	Normal ZZ	-2.217E-04	2.22E-04
	Shear XY	-4.7E-05	4.564E-05
	Shear YZ	-9.557E-05	9.818E-05
	Shear XZ	-1.339E-04	1.663E-04
Contact Pressure			
	Total	0 MPa	0.876 MPa
	X	-0.171 MPa	0.155 MPa
	Y	-0.343 MPa	0.343 MPa
	Z	-0.86 MPa	0.857 MPa
Contact Force			
	Total	0 N	0.257 N
	X	-0.126 N	0.136 N
	Y	-0.254 N	0.246 N
	Z	-0.133 N	0.132 N

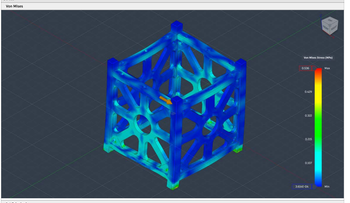
Safety Factor

Safety Factor

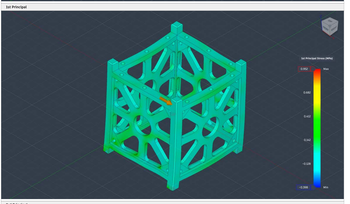


Stress

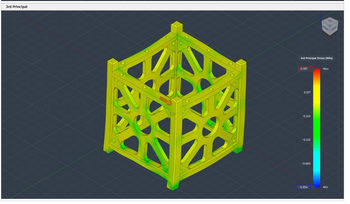
Von Mises

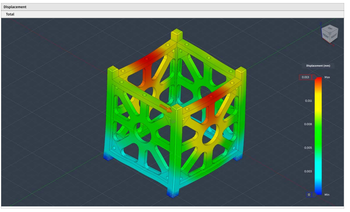


1st Principal



3rd Principal



Displacement

Total